IAL Physics Unit 2 — Drift Velocity (I = nqvA) | Recall Pack

IAL Physics Unit 2 — Drift Velocity & the Transport Equation $I = nqvA$

The whole topic in one mental model. A current is a *crowd* of charge carriers shuffling slowly down a wire. Picture a cylinder of conductor: in a time Δt every carrier drifts forward by $v\,\Delta t$, so a slab of volume $Av\,\Delta t$ sweeps past any cross-section. That slab holds $nAv\,\Delta t$ carriers, each carrying charge q, so the charge crossing per second is $I = nqvA$. One equation contains everything: n (how many carriers there are), q (charge each carries), v (how fast they drift), A (how wide the pipe is). The drift is astonishingly slow (~ 0.1 mm s^{-1} in copper) yet lamps light instantly — because the *electric field*, not the electrons, races down the wire at almost the speed of light and starts the whole crowd moving at once (the marble-tube picture). Rearranged, $v = I/(nqA)$, so for a fixed current a *narrower* wire (small A) or a material with *fewer* carriers (small n, e.g. a semiconductor) forces a *faster* drift.

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1. Jargon glossary

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Charge carrier	A tiny charged particle that can actually move through the material and carry the current — the "messenger" of charge. In metals these are loose electrons; in liquids/gases they are ions.	A mobile charged particle free to move under an applied electric field: delocalised (free) electrons in metals, positive/negative ions in electrolytes and gases, electrons and holes in semiconductors.	(the particle); charge q in C	NOT the fixed positive ion lattice (those stay put); NOT a property of the circuit; NOT "the current itself".	Textbook Fig 7.8: a metallised sphere physically ferries charge across plates — a visible "charge carrier". Metals: $\approx$ one free electron per atom.	**Common trap:** "Resistors store electrons" is FALSE — carriers pass *through* a resistor; the resistor turns electrical energy into heat via collisions, it does not hold the carriers.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Drift velocity v	The slow, steady crawl speed of the carriers *along* the wire once a voltage is applied — the average net forward speed, on top of their wild random jiggling. Typically less than a millimetre per second.	The mean (average) velocity of charge carriers in the direction of conventional current flow when a p.d. is applied. The directed component left over after the random thermal velocities cancel.	v; m s^{-1}	NOT the random thermal speed ($\sim 10^6$ m s^{-1}, which cancels to zero net); NOT the speed of the electric field/signal ($\sim c$); NOT a fixed material constant — it depends on I, A, n.	Copper, $I = 0.25$ A, $A = 0.50$ mm^2: $v \approx 3.7 \times 10^{-5}$ m s^{-1} — an electron takes ~ 22 h to cross 3 m.	**Common trap:** confusing drift velocity (slow, the electrons) with how fast the lamp lights (fast, the field). They are different speeds by $\sim 10^{13}$.
Number density n	How *crowded* the carriers are — how many free carriers are packed into each cubic metre of material. Big n = great conductor; $n \approx 0$ = insulator.	The number of free charge carriers per unit volume of the material. Set by the material (and, for semiconductors, by temperature).	n; m^{-3}	NOT "per atom" and NOT "per mm^3"; NOT the current; NOT temperature-independent for semiconductors.	Copper $n \approx 8 \times 10^{28}$ m^{-3}; tungsten $\approx 4 \times 10^{28}$; germanium $\approx 10^{19}$; silicon $\approx 10^{17}$; glass ≈ 0.	**Phuc trap:** n is per cubic metre (m^{-3}), not per mm^3. If a value is quoted in mm^{-3}, convert: 1 mm^{-3} $=$ 10^9 m^{-3}. The OneNote "copper conducting" problem quotes "10^{29} per mm^3" loosely — read it as per m^3.
Carrier charge q	The amount of charge each individual carrier hauls. For an electron that's the elementary charge e.	The charge per carrier. For electrons $q = e = 1.60 \times 10^{-19}$ C (use the magnitude for speed). For doubly-charged ions $q = 2e$.	q; coulombs C; $e = 1.60 \times 10^{-19}$ C	NOT a variable you solve for in metal problems (it is given/known); NOT zero; the *sign* is negative for electrons but you use magnitude for drift speed.	Given on the formula sheet as $q_e = 1.60 \times 10^{-19}$ C; used in every worked example.	**Common trap:** forgetting that a *negative* v from $I = nqvA$ just means the carriers move *opposite* to conventional current (i.e. they are electrons) — it is not an error.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Cross-sectional area A	How wide the "pipe" the carriers flow through is — the area of the slice you'd cut straight across the wire.	The area of the conductor perpendicular to the current. For a circular wire $A = \pi r^2 = \pi d^2/4$. Set by the wire geometry, not the material.	A; m^2	NOT the diameter or radius directly (you must square); NOT a material property; NOT left in mm^2 when substituting.	Tungsten filament $d = 0.025$ mm $\Rightarrow$ $A \approx 4.9 \times 10^{-10}$ m^2; copper lead $d = 0.72$ mm $\Rightarrow A \approx 4.1 \times 10^{-7}$ m^2.	**Phuc trap:** area must be in m^2. 1 mm$^2 =$ $(10^{-3})^2$ m$^2 =$ 1×10^{-6} m^2 — NOT 10^{-3} m^2. If a *diameter* is given, compute $A = \pi(d/2)^2$ first; never plug the diameter in as if it were the area.
Conventional current	The direction we *pretend* the current flows: from + to − outside the cell, as if positive charges moved. A historical convention.	The flow direction defined as the direction a positive charge would move: from the positive terminal to the negative terminal in the external circuit.	I; amperes A	NOT the actual electron direction (electrons go the *opposite* way, toward +); NOT wrong — just a sign convention predating the discovery of the electron.	Textbook Fig 7.9: electrons drift anticlockwise, so conventional current is clockwise.	**Common trap:** drawing electrons and conventional current in the same direction. In a metal the electrons drift *toward the positive terminal*, opposite to I.
Thermal (random) motion	The frantic, directionless zig-zagging of free electrons even with no battery connected — very fast, but going nowhere on average.	The random high-speed motion of free electrons due to temperature ($\sim 10^6$ m s^{-1}), with no net displacement. Drift is a tiny directed bias added on top when a field is applied.	$\sim 10^6$ m s^{-1} (random)	NOT the drift velocity; NOT a current on its own (it averages to zero); NOT caused by the battery.	Textbook 7.4: electrons jiggle "at speeds approaching one-thousandth the speed of light", colliding with the lattice.	**Phuc trap (lesson 03:18–06:05):** Phuc asked "if there's no circuit, is there still electricity in the wire?" Yes — the wire is full of electrons in random thermal motion, but with no p.d. there is *no net drift* and so no current.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Instant-on (the field, not the electron)	Why the light comes on the moment you flick the switch even though electrons crawl: the electric field — the "push" — fills the whole wire at almost light speed and starts every electron drifting at once.	When the switch closes, the electric field is established along the entire conductor at $\approx 3 \times 10^8$ m s^{-1}, so all carriers begin drifting almost simultaneously. With $n \sim 10^{28}$ m^{-3}, the charge-per-second $I = nqvA$ is large despite tiny v.	field speed $\approx c = 3 \times 10^8$ m s^{-1}	NOT the electrons travelling fast; NOT signal carried *by* one electron moving end-to-end.	Marble-tube analogy: push one marble in, one pops out the far end immediately, though each marble barely moves.	**Phuc trap (lesson, "Is it the same Charlie?"):** Phuc asked whether the *same* electron that leaves the battery reaches the lamp. No — it's the marble-tube picture: a carrier exits one end as another enters; the *field* propagates, not one tagged electron.
Metal vs semiconductor vs insulator	What decides if something conducts: how many free carriers it has. Metals are crammed with them, semiconductors have far fewer, insulators have essentially none.	Conduction is governed by n (since q, A, I are circuit/given): metals $n \sim 10^{28}$–10^{29} m^{-3}; semiconductors $n \sim 10^{16}$–10^{19} m^{-3}; insulators $n \approx 0$. Heating a semiconductor frees more carriers, raising n and lowering resistance.	n ranges (m^{-3})	NOT a difference in q or A; NOT fixed for semiconductors (temperature-dependent).	Copper $\sim 8 \times 10^{28}$; germanium $\sim 10^{19}$; silicon $\sim 10^{17}$; glass ≈ 0.	**Common trap:** thinking semiconductors are slow because they "conduct badly". For the *same* current and area a semiconductor has a *faster* drift v (because n is tiny) — $v = I/(nqA)$.

Continuity of current	In a single loop the current is the same everywhere, so where the wire gets thinner the carriers must speed up to keep the same number flowing past per second.	Conservation of charge / Kirchhoff's current law: I is identical at every point of a series circuit. With I and n fixed, $v \propto 1/A$ — a narrower section means a higher drift velocity.	I same; $v \propto 1/A$	NOT "current is used up"; NOT a change in n or q at the constriction.	Tungsten filament (A small) has $v \sim 1600\times$ the copper leads' drift, same I (textbook 160 mA example).	**Common trap:** thinking the thin filament carries *less* current. It carries the *same* current — that is exactly why its drift velocity is so much higher.

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2. Formula / case table

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Transport equation	$I = nqvA$	I = current; n = number density; q = carrier charge; v = drift velocity; A = cross-section	I A; n m^{-3}; q C; v m s^{-1}; A m^2	The master equation for any charge-carrier / drift-velocity question.	$v = \dfrac{I}{nqA}$; $n = \dfrac{I}{qvA}$; $A = \dfrac{I}{nqv}$; $q = \dfrac{I}{nvA}$	Order of factors is irrelevant ($I = nAvq = nqvA$); but every symbol must be in SI. Mnemonic: "**n**aughty **q**uirky **v**ampires **A**te".	Copper, $n = 6.4 \times 10^{28}$, $q = 1.6 \times 10^{-19}$, $v = 1 \times 10^{-3}$, $A = 1 \times 10^{-6}$: $I = 6.4 \times 1.6 \times 10^{0} \approx 10$ A (OneNote Q1).
Drift velocity (rearranged)	$v = \dfrac{I}{nqA}$	as above; solving for the drift speed	m s^{-1}	The single most common task: "find the drift velocity".	$I = nqvA$; $A = \dfrac{I}{nqv}$	Do it in *one* step — no need to find ΔQ or ΔN first. Keep everything in SI before dividing.	$I = 0.5$ A, $n = 6.4 \times 10^{28}$, $q = 1.6 \times 10^{-19}$, $A = 2 \times 10^{-6}$: $v = 0.5/(2.05 \times 10^{4}) \approx 2.4 \times 10^{-5}$ m s^{-1} (OneNote Q2).

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Swept-volume derivation	$\Delta V = Av\,\Delta t,\quad \Delta N = nAv\,\Delta t,\quad \Delta Q = nqAv\,\Delta t \Rightarrow I = \dfrac{\Delta Q}{\Delta t} = nqvA$	ΔV = volume swept in Δt; ΔN = carriers in it; ΔQ = their charge	(SI throughout)	Deriving / explaining *where* $I = nqvA$ comes from (not examined as a derivation, but builds understanding).	each step divides by Δt at the end	You are "not asked to derive this in the exam" (textbook tip) — but knowing it stops symbol confusion.	In Δt a carrier moves $v\Delta t$; slab $Av\Delta t$ holds $nAv\Delta t$ carriers $\Rightarrow$ charge $nqAv\Delta t$ per Δt = $nqvA$.
Circular cross-section	$A = \pi r^2 = \dfrac{\pi d^2}{4}$	r = radius; d = diameter	A m^2; r, d m	Whenever a *diameter* (or radius) is given instead of an area.	$r = \sqrt{A/\pi}$; $d = 2\sqrt{A/\pi}$	**Phuc trap:** compute A *first*, in m^2. Don't plug d in as if it were A, and don't forget to halve the diameter.	$d = 0.72$ mm: $A = \pi(0.36 \times 10^{-3})^2 \approx 4.07 \times 10^{-7}$ m^2 (copper lead).
Area unit conversion	$1\,\text{mm}^2 = 1 \times 10^{-6}\,\text{m}^2$; $1\,\text{cm}^2 = 1 \times 10^{-4}\,\text{m}^2$	converting quoted areas to SI	m^2	Almost every numerical question quotes mm^2 or cm^2.	$1\,\text{m}^2 = 10^6\,\text{mm}^2$	**Phuc trap:** $1\,\text{mm}^2 \neq 10^{-3}\,\text{m}^2$. Square the length factor: $(10^{-3})^2 = 10^{-6}$.	$2.0\,\text{mm}^2 = 2.0 \times 10^{-6}\,\text{m}^2$ before substituting (self-test Q2).
Number-density conversion	$1\,\text{mm}^{-3} = 1 \times 10^9\,\text{m}^{-3}$	converting a per-mm^3 density to SI	m^{-3}	When n is (sloppily) quoted per mm^3.	$1\,\text{m}^{-3} = 10^{-9}\,\text{mm}^{-3}$	**Phuc trap:** n for copper is $\sim 8 \times 10^{28}$ **m**$^{-3}$; the OneNote "10^{29} per mm^3" is loose wording — treat n as per m^3.	$5\,\text{mm}^{-3} = 5 \times 10^9\,\text{m}^{-3}$.
Continuity (proportionality)	$v \propto \dfrac{1}{A}$ (same I, n, q)	v rises as A falls, for fixed current and material	—	Comparing drift speed in thick vs thin sections of one series circuit.	$\dfrac{v_1}{v_2} = \dfrac{A_2}{A_1}$	Same I everywhere in series — the current is **not** shared or used up. Smaller $A \Rightarrow$ larger v.	Wire Y has $2\times$ the diameter of $X \Rightarrow A_Y = 4A_X \Rightarrow v_X : v_Y = 4 : 1$ (self-test Q4).

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Material comparison (via n)	$v = \frac{I}{nqA}$, so $v \propto \frac{1}{n}$ (same I, A, q)	smaller $n \Rightarrow$ larger v	m s^{-1}	Explaining why a semiconductor's carriers drift faster than a metal's at the same I, A.	$\frac{v_{\text{semi}}}{v_{\text{metal}}} = \frac{n_{\text{metal}}}{n_{\text{semi}}}$	Metals conduct *better* yet have *slower* drift — because they have far more carriers (n huge).	Copper $n \sim 8 \times 10^{28}$ vs germanium $\sim 10^{19}$: $\sim 10^9\times$ faster drift in germanium, same I, A.
Carriers per second	$\frac{\text{electrons}}{\text{s}} = \frac{I}{q}$	rate at which carriers cross a section	s^{-1}	"How many electrons pass per second?" type parts.	$I = q \times (\text{rate})$	This is I/q, *not* $I/(nqA)$ — different question (rate of carriers vs their speed).	$I = 10$ A: rate $= 10/1.6 \times 10^{-19} \approx 6.25 \times 10^{19}$ electrons s^{-1} (OneNote extended Q).
Ohm's law (sanity check)	$V = IR \Rightarrow R = \frac{V}{I}$	V p.d.; I current; R resistance	V, A, Ω	When a drift problem also asks for R or current from p.d.	$I = \frac{V}{R}$; $R = \frac{V}{I}$	**Phuc trap:** $R = V/I$, NOT $R = IV$. Dividing V by I gives resistance; *multiplying* gives power $P = IV$.	$V = 2$ V, $I = 0.5$ A $\Rightarrow R = 2/0.5 = 4\,\Omega$ (not $1\,\Omega$).
Time to traverse a length	$t = \frac{L}{v}$	L = wire length; v = drift velocity	s	"How long for an electron to travel from switch to lamp?" — and contrasting with field speed L/c.	$v = L/t$; $L = vt$	The electron time is huge ($\sim$ hours); the *field* time is $L/c \sim$ nanoseconds — that's why the lamp lights "instantly".	$L = 3$ m, $v = 3.7 \times 10^{-5}$: $t = 3/3.7 \times 10^{-5} \approx 8.1 \times 10^4$ s ≈ 22 h (self-test Q8 / textbook Q9).

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3. Magnitude benchmarks — one-glance table

Situation	n (m^{-3})	A	I	Drift v	Take-away
Copper wire, 1 mm^2, 1 A	$\sim 8.5 \times 10^{28}$	1×10^{-6} m^2	1 A	~ 0.07 mm s^{-1}	Metal drift is sub-mm/s — slower than a snail.
Copper wire, 1 mm^2, 10 A	$\sim 8.5 \times 10^{28}$	1×10^{-6} m^2	10 A	~ 0.7 mm s^{-1}	$v \propto I$: ten times the current, ten times the drift.
Tungsten filament, 160 mA	4.0×10^{28}	4.9×10^{-10} m^2	0.16 A	~ 51 mm s^{-1}	Thin $A \Rightarrow$ fast drift, $\sim 1600\times$ the leads.
Semiconductor strip, same I	$\sim 7 \times 10^{22}$	(mm-scale)	10 mA	~ 0.3 m s^{-1}	Tiny $n \Rightarrow$ drift $\sim 10^3$-$10^9\times$ faster than metal.
Electric field / signal	—	—	—	$\approx 3 \times 10^8$ m s^{-1}	The field, not the electron, makes the lamp light instantly.

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4. Prefix cheat-sheet (no excuses)

Prefix	Symbol	Multiplier
nano	n	10^{-9}
micro	**µ**	$\mathbf{10^{-6}}$ (six zeros)
milli	**m**	$\mathbf{10^{-3}}$ (three zeros)
kilo	k	10^{+3}
mega	M	10^{+6}

Phuc trap (lesson ≈15:03): Phuc wrote μA $= 10^{-3}$ — wrong. **micro** $= 10^{-6}$**, milli** $= 10^{-3}$**.** Mixing them makes an answer $1000\times$ too large. Write this table at the top of every paper.

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5. Big picture (last-minute recall)

- $I = nqvA$ — learn the order ("naughty quirky vampires Ate"): current = (how many)(charge each)(how fast)(how wide).
- Drift is *slow* (mm s^{-1} or less); the *field* is *fast* ($\approx c$). The lamp lights instantly because of the field, not the electrons — marble-tube picture.
- $v = I/(nqA)$: same I, smaller $A \Rightarrow$ faster drift (continuity); smaller $n \Rightarrow$ faster drift (semiconductor vs metal).
- Big n = good conductor. Metals $\sim 10^{28}$-10^{29}; semiconductors $\sim 10^{16}$-10^{19}; insulators $n \approx 0 \Rightarrow I = 0$ whatever the voltage.
- Always convert to SI first: $1\,$mm$^2 = 10^{-6}\,$m^2; if given a diameter, $A = \pi(d/2)^2$.

- micro $= 10^{-6}$, milli $= 10^{-3}$ — never confuse them.
- A negative v just means carriers move opposite to conventional current (electrons) — not an error.
- Electrons in a metal drift *toward the + terminal*, opposite to conventional current.